Build your own object detection system using YOLO

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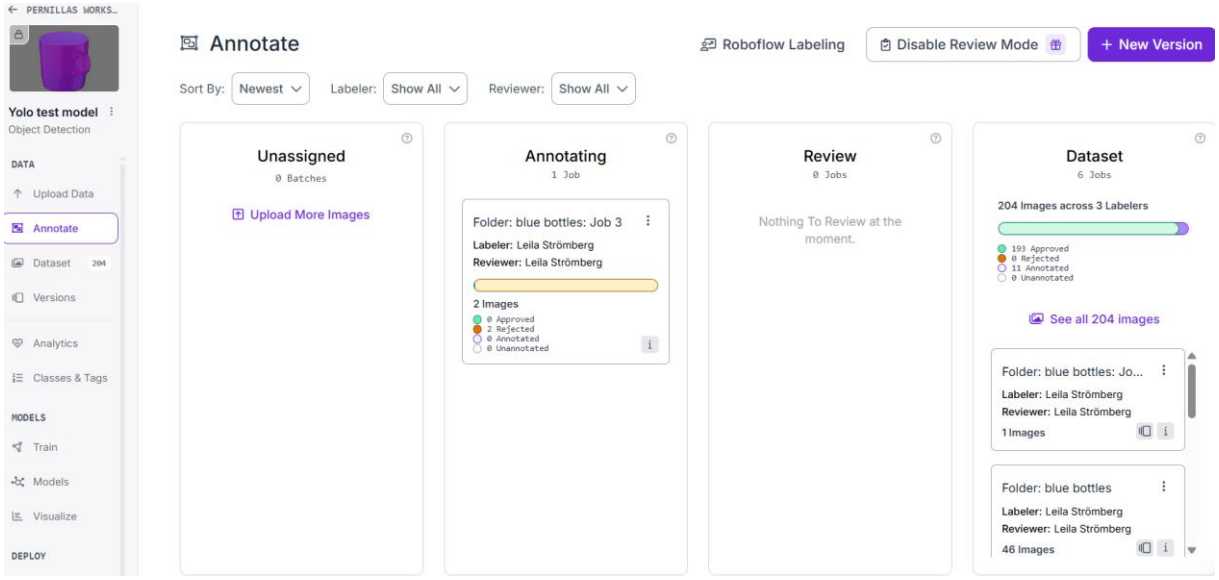
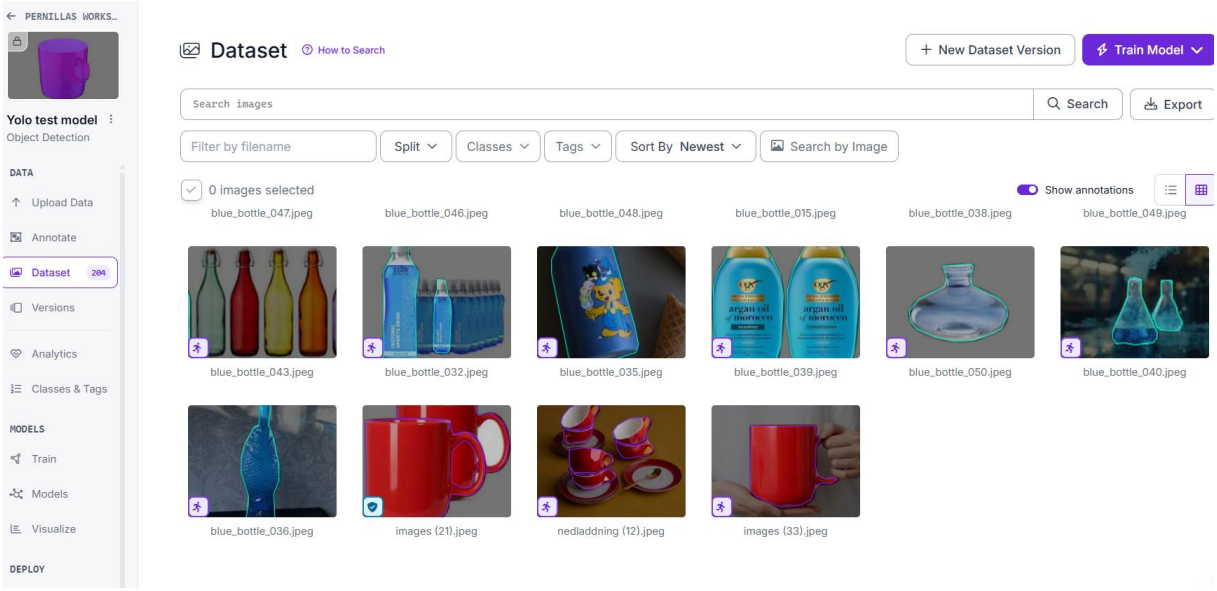
Objective: Make an own model that detect three object types: a red cup, a blue bottle, and a phone.

Creating a dataset.

We used Roboflow to create a dataset. First we collected around 160 pictures. We took some pictures ourselves, we used a set of pictures found on Roboflow and we used google picture search to find some more. All these pictures were uploaded to Roboflow were we annotated them with class labels:





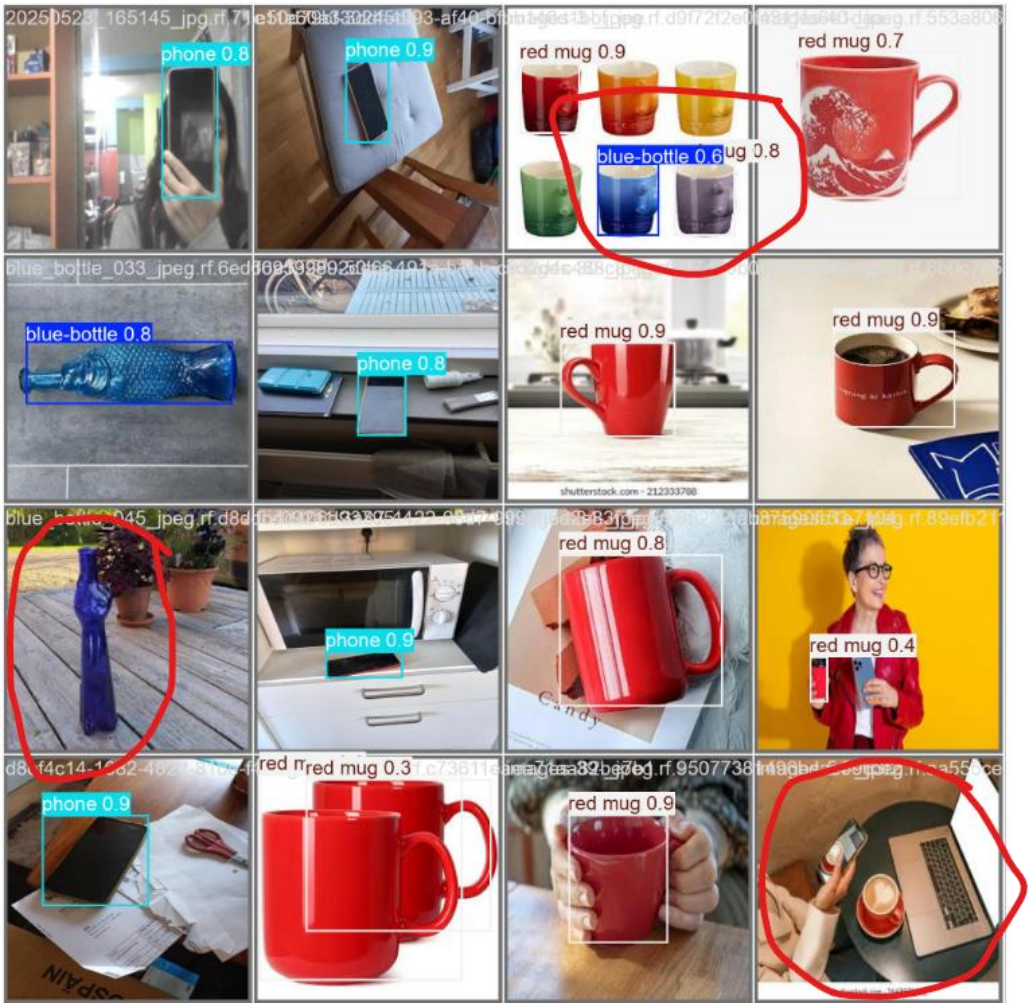
From roboflow we could create a dataset with some augmented versions (flipped, rotaded) of the pictures to expand the dataset. The advantage with roboflow is that the dataset is available online and we could use it in colab.

Models created.

We used colab since our computers are too slow for big modelling. Both of us are new to colab so the notebook ended up in Leilas titanic Git repository:

[titanic-capstone-project/notebooks/train-yolov8-object-detection-on-custom-dataset.ipynb at main · hashtag-squirrel/titanic-capstone-project]

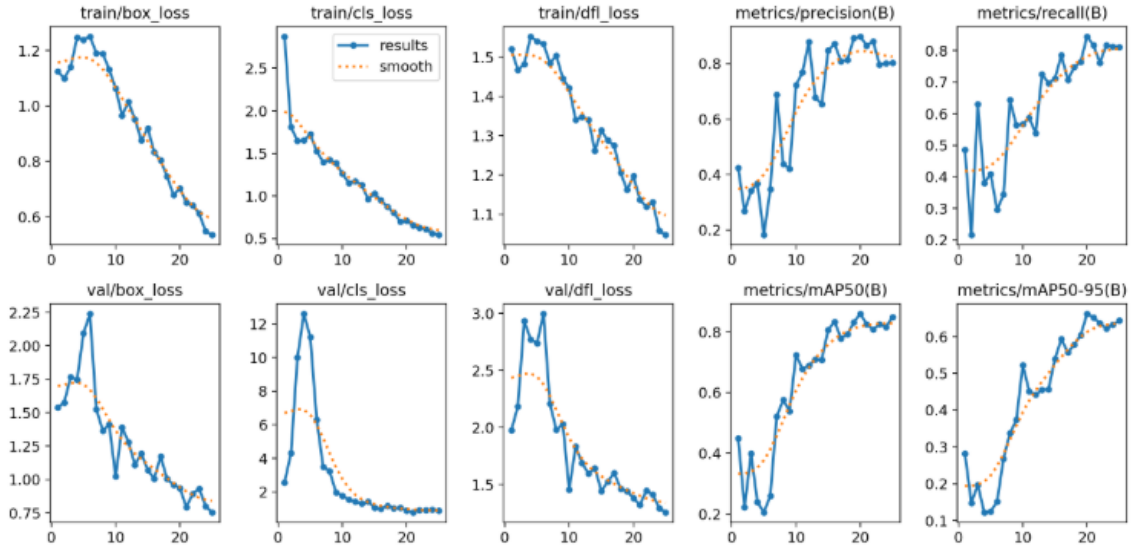
We started with 25 epochs. This model was fairly good at predicting pictures but makes some mistakes:



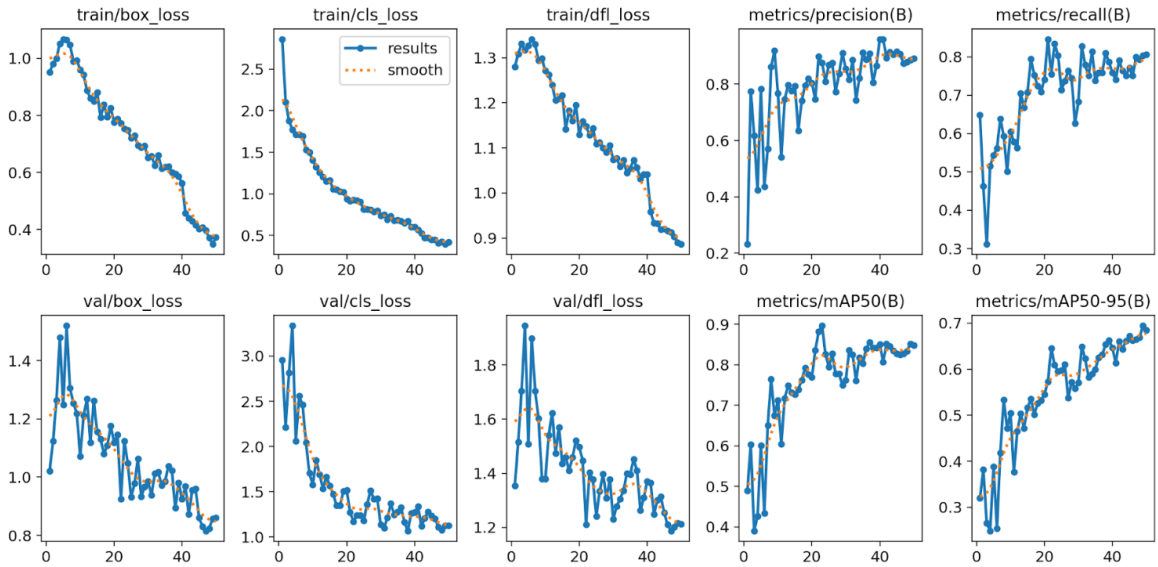
If we look at the loss curves we see that it probably still can improve:

```
%cd {HOME}
Image(filename=f'{HOME}/runs/detect/train3/results.png', width=600)
```

```
/content/{HOME}/datasets
```

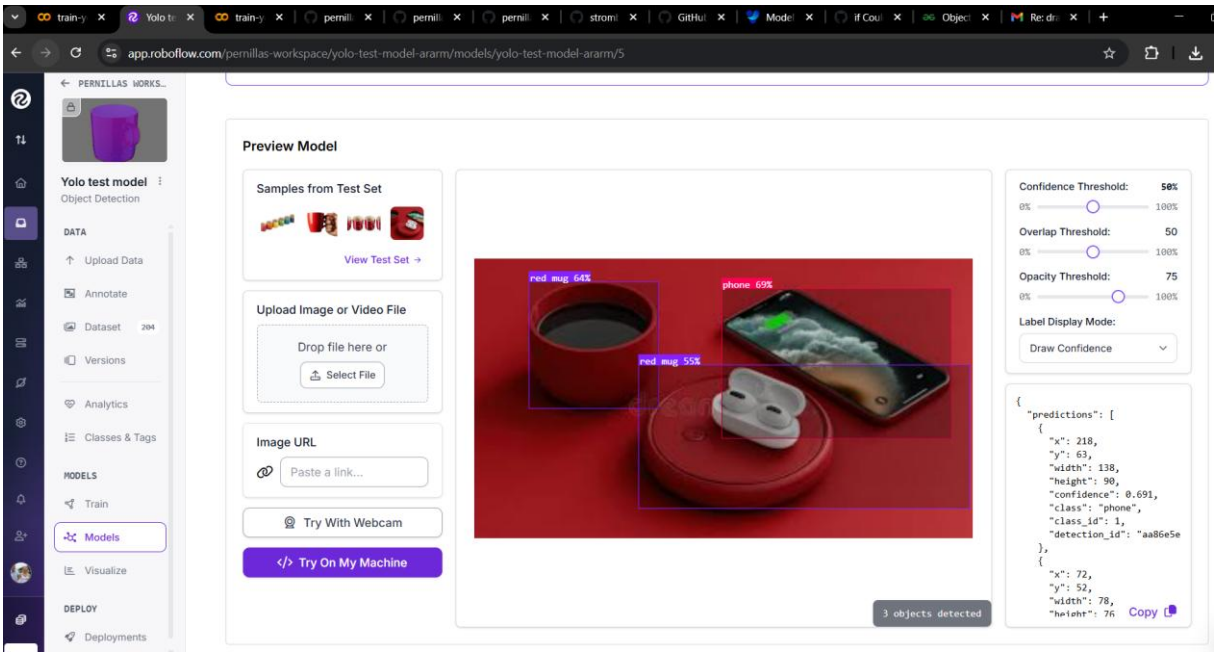


We doubled the epochs to 50 to see if we could improve the model further (and downsized imagesize a bit to not take too long to train).



But when we look at the test pictures it still makes mistakes. To improve the model further we would need to add more learning pictures.

The trained model could be saved back to Roboflow. But this is a trial version for 2 weeks only so

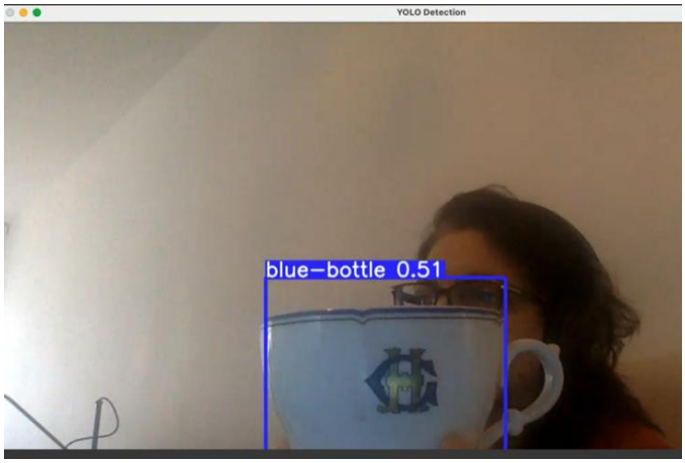


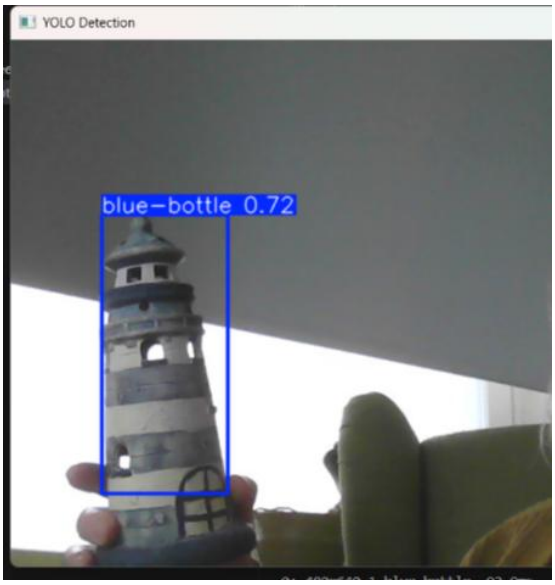
we saved the best.pt of the model for the next part of the lab.

Live feed.

The last part of the lab was to use the model in live feed to identify what is in the frame.

We used or trained model, best.pt, and it could identify correctly mobile phones and cups. It seems to rely heavily on colour. A white-blueish cup would be guessed as blue bottle rather than red mug.





Git repository (video capture)

[strombergleila-source/object_detection]

References:

[How to Train YOLOv8 Object Detection on a Custom Dataset]